MRI Brain Age Data - Insight 46 Phase 1

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Document details

Categories of variables	MRI Brain Age Data for Insight 46 Phase 1 (Generated using BrainAgeR software version 1.0)
Summary of work undertaken	Cleaning of data: 1. Selection of Phase 1 data 2. Checking for duplicates 3. Identifying and accounting for missing data
Names of source variables	Participant, ID, Date, of, birth, Baseline, scan, date, Baseline, age, at, scan, dob)/365.25, Unadjusted, baseline, brain-predicted, age, Adjusted, baseline, brain-predicted, age, Adjusted, baseline, Brain-PAD, bl_bpa_adjusted, bl_age_at_scan, Follow, up, scan, date, up, age, at, scan, dob)/365.25, follow, up, brain-predicted, age, Adjusted, follow, up, brain-predicted, age, follow, up, Brain-PAD, y2_bpa_adjusted, y2_age_at_scan, Is, participant, part, of, datafreeze?, participant, who, is, in, freeze, doesn't, have, a, brain-predicted, age
Output variables	brn_mri_i46p1, brn_age_i46p1, brn_bpaun_i46p1, brn_bpa_i46p1, brn_pad_i46p1
Papers using these variables	Wagen, A. Z., et al. (2022). "Life course, genetic, and neuropathological associations with brain age in the 1946 British Birth Cohort: a population-based study." Lancet Healthy Longev **3**(9): e607-e616.

Prior Derivation (Previously performed by W. Coath)

Brain age values were calculated on the UCL Centre for Medical Image Computing Computer Science cluster using version 1 of the 'BrainAgeR' software (Cole et al., 2019 doi: 10.31219/osf.io/3b6zu).

Brain-predicted age was produced for all Insight 46 Phase 1 participants with a T1 MRI scan which passed quality control.

The **brn_bpa_i46p1** variable is the brain-predicted age after adjustment for age bias. The age bias adjustment was performed by subtracting 3.33 from the unadjusted brain age and dividing by 0.91. These values were the intercept and slope of the model in the training dataset of brain-predicted age regressed on chronological age.

Brain-predicted age difference (**Brain-PAD**) is provided for the adjusted brain-predicted age estimates only. This was calculated by subtracting chronological age from adjusted brain-predicted age.

Cleaning

Duplicates: None

Missing data: Missing data recorded as "NA"

Missing data for 34 participants.

1. This includes n = 31 participants who attended an Insight 46 Phase 1 visit but did not undergo an MRI scan (Most due to claustrophobia, some due to PET-MRI incompatibility, recent illness and withdrawal before being rescheduled).
2. The remaining participants had an MRI scan during their Insight 46 Phase 1 visit but it did not pass quality control

Variables in this document

5 high-confidence matches

Variable	Label	How it was found
Sub-Studies [57] › Insight46 [699] › Brain MRI [6033] › Brain age [60335]		
brn_age_i46p1	Age at brain scan (years) - Cross-sectional - Insight 46 Phase 1	topsheet field
brn_bpa_i46p1	Adjusted baseline brain-predicted age - Cross-sectional - Insight 46 Phase 1	topsheet field
brn_bpaun_i46p1	Unadjusted baseline brain-predicted age - Cross-sectional - Insight 46 Phase 1	topsheet field
brn_mri_i46p1	Did the participant have a brain scan - Cross-sectional - Insight 46 Phase 1	topsheet field
brn_pad_i46p1	Adjusted baseline brain predicted age differeince (Brain-PAD) - Cross-sectional - Insight 46 Phase 1	topsheet field